

Rational - Canonical Form

Jordan - Canonical Form

Smith - Canonical Form.

Positive form

bilinear form

Matrix Analysis, (Matrix Norm)

Determinant.

Fourier series

Integral and differential form

More topics in series and sequence of functions.

Differential Equation.

Goal

1. All Problem in:

Drumitt,

Hoffman,

Rudin, KKG

Munkres,

Hatcher,

O'neil,

Silverman.

Solve, and optimize in my own language.

Every study papers should be written under the fixed rules:

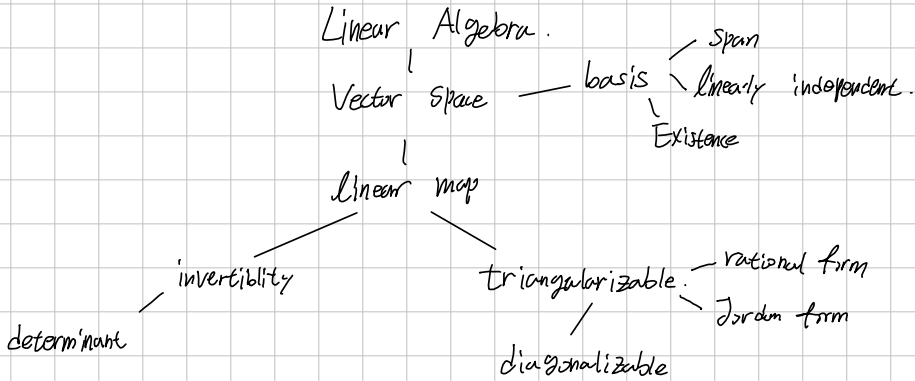
Prop: Proposition. A useful fact

Lemma: Lemma.

Thm: Theorem. A main result in the topic.

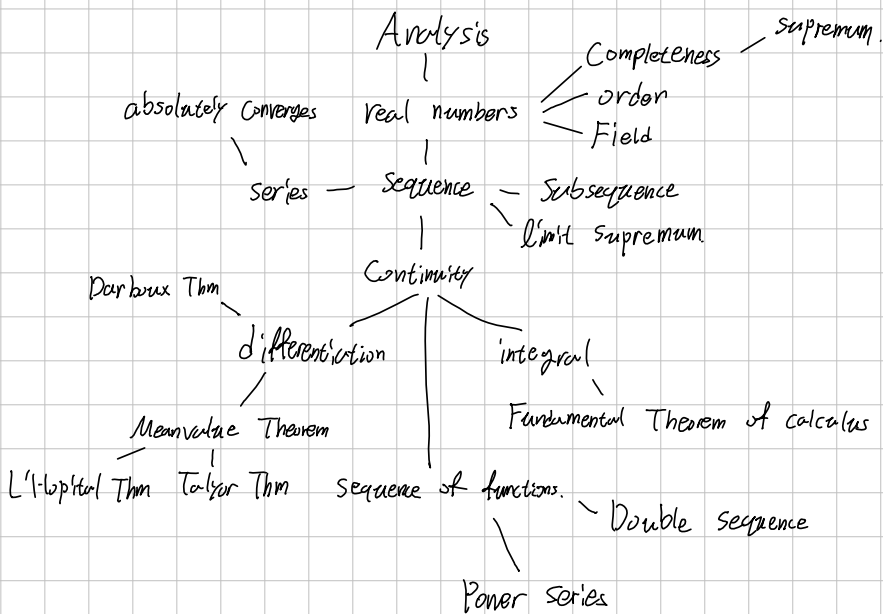
Cor.: Corollary.

Main Process to Understand:



Linear Algebra - View of Geometry

Inner Product.





Topology

Complex Analysis

Complex number  $\begin{cases} \text{Completeness} \\ \text{Field} \end{cases}$

Complex function

differentiation

Series representation

integral

Cauchy-Riemann Equation.

Cauchy integral formula

analytic

# Algebraic Topology



# Differential Geometry

